Elementary Analysis

Section 35: Riemann–Stieltjes Integrals

Exercises 35.1–35.12, with every subpart
Detailed worked solutions

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Contents

35 Riemann–Stieltjes integrals **2**

35.1 Exercise 35.1: Why an increasing function has one-sided limits 2
35.2 Exercise 35.2: Separate a continuous increase from a jump 3
35.3 Exercise 35.3: Integrating against the integer step function 3
35.4 Exercise 35.4: A smooth trigonometric integrator 4
35.5 Exercise 35.5: The rational indicator sees continuous and atomic mass differently . 4
35.6 Exercise 35.6: Uniform limits preserve Stieltjes integrability 5
35.7 Exercise 35.7: Products, squares, maxima, and minima 6
35.8 Exercise 35.8: A continuous density defines the weight 7
35.9 Exercise 35.9: A weighted mean value is attained 7
35.10 Exercise 35.10: A polynomial integrated against itself 8
35.11 Exercise 35.11: Change variables without differentiating the substitution 9
35.12 Exercise 35.12: A strictly increasing function with dense jumps 9

35 Riemann–Stieltjes integrals

Read this before calculating. An ordinary integral weights an interval by its length. A Stieltjes integral weights it by the increase of an increasing function F. Here "increasing" permits equality: $s < t$ implies $F(s) \le F(t)$. Ross uses a modified definition that handles jumps explicitly. This matters: a formula from another book's traditional Stieltjes theory may not apply to this section.

Write $F(t-)$ and $F(t+)$ for the one-sided limits. On the particular interval $[a, b]$ being integrated over, the endpoint conventions are $F(a-) = F(a)$ and $F(b+) = F(b)$. Thus the mass at t is $\Delta F(t) = F(t+) - F(t-)$. For a partition $P = \{a = t_0 < \cdots < t_n = b\}$ set

$$w_k = F(t_k-) - F(t_{k-1}+), \qquad J_F(f, P) = \sum_{k=0}^{n} f(t_k)\Delta F(t_k).$$

The upper sum is $J_F(f, P) + \sum_{k=1}^{n} (\sup_{(t_{k-1}, t_k)} f) w_k$; the lower sum uses infima instead. The numbers w_k and $\Delta F(t_k)$ are nonnegative, and together they add to $W = F(b) - F(a)$. In particular, if $|h| \le B$, then every upper and lower sum for h lies between $-BW$ and BW.

The essential test is Theorem 35.6: a bounded function is F-integrable precisely when, for every $\varepsilon > 0$, some partition satisfies $U_F(f, P) - L_F(f, P) < \varepsilon$. The jump terms cancel in this difference. We will often use the equivalent expression

$$U_F(f, P) - L_F(f, P) = \sum_{k=1}^{n} \operatorname{osc}(f, (t_{k-1}, t_k)) w_k,$$

where $\operatorname{osc}(f, I) = \sup_I f - \inf_I f$ is the oscillation of f on I.

35.1 Exercise 35.1: Why an increasing function has one-sided limits

What we must show. Identify both one-sided limits of F by a supremum or an infimum, including the indicated endpoints.

Plan. Increasing values cannot move back and forth. Their supremum on the left, or infimum on the right, is therefore the only possible limiting value. We verify this using the definition of a limit.

(a) Approach $t \in (a, b]$ from the left.

The set $A = \{F(x) : a < x < t\}$ is nonempty and bounded above by $F(t)$. Consequently $L = \sup A$ is a finite real number. Given $\varepsilon > 0$, the definition of supremum supplies $y \in (a, t)$ such that $L - \varepsilon < F(y) \le L$. For every x with $y < x < t$, monotonicity gives

$$L - \varepsilon < F(y) \le F(x) \le L.$$

Take $\delta = t - y > 0$. If $x \in [a, b]$ and $0 < t - x < \delta$, then $y < x < t$, so $|F(x) - L| < \varepsilon$. This is exactly $\lim_{x \to t-} F(x) = L$. The proof works when $t = b$, since it only uses points to the left of t.

(b) Approach $t \in [a, b)$ from the right.

The nonempty set $B = \{F(x) : t < x < b\}$ is bounded below by $F(t)$. Put $R = \inf B$. Given $\varepsilon > 0$, choose $y \in (t, b)$ with $R \le F(y) < R + \varepsilon$. Whenever $t < x < y$,

$$R \le F(x) \le F(y) < R + \varepsilon.$$

Now $\delta = y - t$ proves $\lim_{x \to t+} F(x) = R$. At $t = a$ the same argument is valid using only points to the right.

Conclusion. The left limit is $\sup\{F(x) : a < x < t\}$ and the right limit is $\inf\{F(x) : t < x < b\}$. Both are finite.

35.2 Exercise 35.2: Separate a continuous increase from a jump

What we must show. Evaluate $\int_0^3 x^2\, dF(x)$ for the piecewise function in Example 4.

Record all three pieces, especially the jump.

The function is $F(t) = 0$ for $t < 0$, $F(t) = t^2$ for $0 \le t < 2$, and $F(t) = t + 5$ for $t \ge 2$. At 2 its left limit is 4 and its right limit is 7, so the jump has size 3. Write $F = F_1 + 3J_2$, where on $[0, 3]$

$$F_1(t) = \begin{cases} t^2, & 0 \le t \le 2, \\ t + 2, & 2 \le t \le 3. \end{cases}$$

This F_1 is continuous. Its derivative is $2t$ on the first piece and 1 on the second. We apply Theorem 35.13 separately on the two pieces; no derivative at the joining point is needed.

Calculate each contribution.

Linearity in the integrator, interval additivity, and the jump rule $\int f\, dJ_2 = f(2)$ give

$$\begin{aligned} \int_0^3 x^2\, dF(x) &= \int_0^2 x^2(2x)\, dx + \int_2^3 x^2\, dx + 3(2^2) \\ &= \left[\frac{x^4}{2}\right]_0^2 + \left[\frac{x^3}{3}\right]_2^3 + 12 \\ &= 8 + \frac{27 - 8}{3} + 12 = \frac{79}{3}. \end{aligned}$$

The extra 12 is essential: differentiating the smooth pieces alone would lose the mass concentrated at 2.

Conclusion. $\displaystyle\int_0^3 x^2\, dF(x) = 79/3.$

35.3 Exercise 35.3: Integrating against the integer step function

What we must show. Evaluate the three requested integrals against $F(t) = \lfloor t \rfloor$.

Plan. This function has jumps of size 1 at the integers. On a specified integration interval we include only the jumps belonging to the restricted function, with Ross's endpoint convention.

(a) Integrate x on $[0, 6]$.

On this interval $F(0-)$ means $F(0) = 0$, rather than the limit of the original function from negative numbers. Also $F(0+) = 0$. There is therefore no mass at 0. There is a mass of 1 at each of $1, 2, 3, 4, 5, 6$. The last one is included because $F(6+) = F(6) = 6$ and $F(6-) = 5$. Example 1 gives

$$\int_0^6 x\, dF(x) = \sum_{j=1}^{6} j = \frac{6 \cdot 7}{2} = 21.$$

(b) Integrate x^2 on $[0, 3]$.

The contributing jumps are at $1, 2, 3$, with the same endpoint reasoning. Thus

$$\int_0^3 x^2\, dF(x) = 1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14.$$

(c) Integrate x^2 on $[1/4, \pi/4]$.

Both endpoints lie strictly between 0 and 1. The integrator is identically 0 on the whole interval; every interval increment and every endpoint jump is zero. Its upper and lower sums are both zero.

Conclusion. The three values are 21, 14, and 0, respectively.

35.4 Exercise 35.4: A smooth trigonometric integrator

What we must show. Integrate x against $F(t) = \sin t$ on the two specified intervals.

Check the conversion to an ordinary integral.

On $[-\pi/2, \pi/2]$, $F'(t) = \cos t \geq 0$, so F is increasing. Both F' and $f(t) = t$ are continuous. Theorem 35.13 therefore permits the replacement $dF(x) = \cos x\, dx$ on each interval.

(a) The interval $[0, \pi/2]$.

Integration by parts with $u = x$ and $dv = \cos x\, dx$ gives

$$\int x \cos x\, dx = x \sin x - \int \sin x\, dx = x \sin x + \cos x.$$

Consequently

$$\int_0^{\pi/2} x\, dF(x) = [x \sin x + \cos x]_0^{\pi/2} = \frac{\pi}{2} - 1.$$

(b) The interval $[-\pi/2, \pi/2]$.

The function $x \cos x$ is odd: replacing x by $-x$ changes its sign. Its integrals over the two halves cancel, so the answer is 0. Alternatively the displayed antiderivative has value $\pi/2$ at both endpoints, giving difference 0.

Conclusion. The values are $\pi/2 - 1$ and 0.

35.5 Exercise 35.5: The rational indicator sees continuous and atomic mass differently

What we must show. For $f = \mathbf{1}_{\mathbb{Q}}$, contrast continuous integrators with the purely discrete integrators in Examples 1 and 3.

(a) A continuous nonconstant integrator.

Every nonempty open interval contains both a rational number and an irrational number. Hence on every partition interval (t_{k-1}, t_k), the supremum of f is 1 and the infimum is 0. Continuity of F

makes all jump terms zero and gives $w_k = F(t_k) - F(t_{k-1})$. Therefore, for every partition,

$$L_F(f, P) = 0, \qquad U_F(f, P) = \sum_{k=1}^{n} [F(t_k) - F(t_{k-1})] = F(b) - F(a).$$

Taking the supremum of the lower sums and infimum of the upper sums changes neither number. Since $F(b) > F(a)$, the lower and upper integrals are different. Thus f is not F-integrable.

(b) A finite or summable collection of point masses.

For $F = \sum_j c_j J_{u_j}$ as in those examples, every bounded function is F-integrable, including this f. Its integral is

$$\int_a^b f \, dF = \sum_j c_j f(u_j) = \sum_{\{j : u_j \in \mathbb{Q}\}} c_j.$$

The formula includes the finite case. In the infinite case it converges absolutely, because $0 \le c_j f(u_j) \le c_j$ and $\sum_j c_j < \infty$. These integrators assign their mass to actual points; the value of f at each mass point is well defined.

Conclusion. For continuous F of positive total increase, f is not integrable. For the discrete F of Examples 1 and 3, it is integrable, with integral equal to the total mass at rational points.

35.6 Exercise 35.6: Uniform limits preserve Stieltjes integrability

What we must show. If each f_n is F-integrable and $f_n \to f$ uniformly on $[a, b]$, prove that f is F-integrable and that the integrals converge.

Plan. Uniform convergence controls every point by the same error. Multiplying that error by the total mass $W = F(b) - F(a)$ controls every upper sum, lower sum, and ultimately the integral.

First establish boundedness.

Take one index N for which $|f(x) - f_N(x)| < 1$ for every $x \in [a, b]$. Such an index exists by uniform convergence. Since f_N is integrable, it is bounded, say $|f_N(x)| \le B$. Then $|f(x)| \le B + 1$ everywhere. This verifies the boundedness required before applying the Darboux criterion.

Make the upper–lower gap small.

Assume $W > 0$; if $W = 0$, every sum and integral is zero and the conclusions are immediate. Fix $\varepsilon > 0$ and select N so that $|f - f_N| \le \eta$ everywhere, where $\eta = \varepsilon/(4W)$. On any nonempty interval I,

$$\sup_I f \le \sup_I f_N + \eta, \qquad \inf_I f \ge \inf_I f_N - \eta.$$

Subtracting gives $\operatorname{osc}(f, I) \le \operatorname{osc}(f_N, I) + 2\eta$. Because f_N is integrable, choose P with $U_F(f_N, P) - L_F(f_N, P) < \varepsilon/2$. Since $\sum_k w_k \le W$,

$$\begin{aligned} U_F(f, P) - L_F(f, P) &\le U_F(f_N, P) - L_F(f_N, P) + 2\eta \sum_k w_k \\ &< \frac{\varepsilon}{2} + 2\eta W = \varepsilon. \end{aligned}$$

Theorem 35.6 proves that f is integrable.

Now estimate the integrals.

Put $\eta_n = \sup_{x\in[a,b]} |f_n(x) - f(x)|$; uniform convergence says $\eta_n \to 0$. The integrable difference $f_n - f$ satisfies $-\eta_n \le f_n - f \le \eta_n$. Order preservation and integration of constants give

$$-\eta_n W \le \int_a^b (f_n - f)\, dF \le \eta_n W.$$

Thus $\left|\int_a^b f_n\, dF - \int_a^b f\, dF\right| \le \eta_n W \to 0$.

Conclusion. Uniform convergence permits passage to the limit under this integral, with error at most the uniform function error times the total mass.

35.7 Exercise 35.7: Products, squares, maxima, and minima

What we must show. Show that all four indicated constructions preserve F-integrability.

(a) Squaring an integrable function.

Choose $B \ge 1$ such that $|f(x)| \le B$ on $[a, b]$. For any two points x, y in the same interval I,

$$\left|f(x)^2 - f(y)^2\right| = |f(x) - f(y)|\,|f(x) + f(y)| \le 2B\,|f(x) - f(y)|\,.$$

The supremum of pairwise differences of a bounded real function is its oscillation. It follows that $\operatorname{osc}(f^2, I) \le 2B \operatorname{osc}(f, I)$. Given $\varepsilon > 0$, take a partition making the gap for f less than $\varepsilon/(2B)$. The gap formula then makes the gap for f^2 less than ε. Since f^2 is bounded, the criterion proves its integrability.

(b) Multiplying two integrable functions.

Linearity makes $f + g$ and $f - g$ integrable. Part (a) makes their squares integrable. The identity

$$fg = \frac{(f+g)^2 - (f-g)^2}{4}$$

then expresses fg as a linear combination of integrable functions. To check the identity, expand both squares: their f^2 and g^2 terms cancel, leaving $4fg$ in the numerator.

(c) Taking the pointwise maximum and minimum.

Theorem 35.10 makes $|f - g|$ integrable. At each point,

$$\max(f, g) = \frac{f + g + |f - g|}{2}, \qquad \min(f, g) = \frac{f + g - |f - g|}{2}.$$

For example, if $f(x) \ge g(x)$ then $|f(x) - g(x)| = f(x) - g(x)$, so the first expression becomes $f(x)$ and the second becomes $g(x)$. The other ordering gives the reversed roles. The formulas and linearity prove both integrability assertions.

Conclusion. f^2, fg, $\max(f, g)$, and $\min(f, g)$ are all F-integrable.

35.8 Exercise 35.8: A continuous density defines the weight

What we must show. For continuous $g \geq 0$ and $F(t) = \int_a^t g(x)\,dx$, show that $\int_a^b f\,dF = \int_a^b f(x)g(x)\,dx$ for continuous f.

Check that F is an admissible increasing integrator.

If $a \leq s < t \leq b$, interval additivity gives

$$F(t) - F(s) = \int_s^t g(x)\,dx \geq 0,$$

because g is nonnegative. Thus F is increasing. The Fundamental Theorem of Calculus, applied to continuous g, gives $F' = g$ on the interior, with the corresponding one-sided derivatives at the endpoints. In particular F' is continuous on the closed interval.

Apply the smooth-integrator formula.

All the hypotheses of Theorem 35.13 are now verified, so

$$\int_a^b f(x)\,dF(x) = \int_a^b f(x)F'(x)\,dx = \int_a^b f(x)g(x)\,dx.$$

If g is identically zero, F is constant. Although the section usually excludes a constant integrator to avoid trivial cases, its sums are still defined and all equal zero; the formula is then also true. If g is nonzero somewhere, continuity and nonnegativity give positive integral, so $F(b) > F(a)$ as in the standing convention.

Conclusion. The density g converts a Stieltjes integral into the usual weighted Riemann integral.

35.9 Exercise 35.9: A weighted mean value is attained

What we must show. Prove the mean-value assertion for continuous f, then recover the weighted integral theorem in Exercise 33.14.

(a) Trap the average between a minimum and a maximum.

The Extreme Value Theorem gives real numbers $m = \min_{[a,b]} f$ and $M = \max_{[a,b]} f$. Put $W = F(b) - F(a) > 0$. The continuous function f is F-integrable by Theorem 35.7. Integrating $m \leq f \leq M$ with order preservation gives

$$mW \leq \int_a^b f\,dF \leq MW.$$

Divide by the positive number W and call the quotient A. We have $m \leq A \leq M$. By the Intermediate Value Theorem, f takes every value between its attained minimum and maximum. Hence some $x \in [a, b]$ satisfies $f(x) = A$. Multiplication by W proves

$$\int_a^b f\,dF = f(x)[F(b) - F(a)].$$

The conclusion naturally allows endpoints: all the mass could be concentrated at a, for example.

(b) Recover the continuous-density theorem.

For continuous $g \geq 0$ set $F(t) = \int_a^t g(s)\,ds$, as in Exercise 35.8. If $g \equiv 0$, both integrals in Exercise 33.14(a) are zero and any $x \in (a, b)$ works. Otherwise $W = \int_a^b g > 0$. Part (a) and Exercise 35.8 give

$$\int_a^b f(t)g(t)\,dt = f(x)\int_a^b g(t)\,dt$$

initially for some $x \in [a, b]$.

Explain why the point can be chosen in the interior here.

Exercise 33.14(a) asks for $x \in (a, b)$, so this small extra point deserves a proof. Let $A = (\int_a^b fg)/(\int_a^b g)$. Suppose no interior point satisfies $f(x) = A$. By continuity and the Intermediate Value Theorem, $f - A$ must then have the same strict sign on all of (a, b). Also some $t_0 \in (a, b)$ has $g(t_0) > 0$: a positive endpoint value of g would imply positive nearby interior values as well. If $f - A > 0$, continuity gives a short interval around t_0 on which both $f - A$ and g have positive lower bounds. Thus $(f - A)g \geq 0$ everywhere and has strictly positive integral, contrary to

$$\int_a^b (f - A)g = \int_a^b fg - A\int_a^b g = 0.$$

If $f - A < 0$, apply the same argument to $(A - f)g$ and obtain the same contradiction. Therefore an interior point does attain A. Taking $g \equiv 1$ recovers the ordinary integral mean-value theorem, as requested in Exercise 33.14(b).

Conclusion. Part (a) is the Stieltjes mean-value formula. With a continuous nonnegative density it gives Exercise 33.14(a), including its interior-point conclusion.

35.10 Exercise 35.10: A polynomial integrated against itself

What we must show. For a positive odd integer n and $F(x) = f(x) = x^n$, evaluate $\int_a^b f\,dF$.

Explain the parity hypothesis.

The map $x \mapsto x^n$ is increasing on all of $\mathbb{R}$ when n is odd. Its derivative is $nx^{n-1} \geq 0$ since $n - 1$ is even. Both f and F' are continuous, so Theorem 35.13 applies on any closed interval $[a, b]$.

Convert and integrate.

We have

$$\begin{aligned}\int_a^b x^n\,d(x^n) = \int_a^b x^n\,nx^{n-1}\,dx &= n\int_a^b x^{2n-1}\,dx \\ &= n\left[\frac{x^{2n}}{2n}\right]_a^b = \frac{1}{2}(b^{2n} - a^{2n}).\end{aligned}$$

The result can be negative if the interval lies on the negative half-line. That is consistent: the weight is nonnegative, but $f(x) = x^n$ is negative there.

Conclusion. $\int_a^b f\,dF = \frac{1}{2}[b^{2n} - a^{2n}]$.

35.11 Exercise 35.11: Change variables without differentiating the substitution

What we must show. For a continuous strictly increasing map $\phi : [c, d] \to [a, b]$ with the stated endpoint values, prove integrability and equality of the two Stieltjes integrals for $g = f \circ \phi$ and $G = F \circ \phi$.

Plan. A partition and its image under ϕ describe exactly the same intervals and mass points. Match their sums term by term. There is no assumption that ϕ is differentiable.

The substitution maps intervals onto intervals.

Continuity, strict increase, and the endpoint values imply that ϕ is a bijection from $[c, d]$ onto $[a, b]$. Its inverse is continuous: for an interior u, points just below and above u have image values strictly below and above $\phi(u)$, which trap the inverse of every sufficiently nearby image point. The same one-sided trapping works at the endpoints. In particular,

$$\phi((u, v)) = (\phi(u), \phi(v)) \qquad (c \le u < v \le d).$$

The composition G is increasing, and g is bounded because f is bounded on the image $[a, b]$.

Match one-sided limits and jumps.

For $u \in (c, d)$, as $v \to u-$ the numbers $\phi(v)$ approach $\phi(u)$ from below. They cover a whole one-sided interval near $\phi(u)$. Therefore

$$G(u-) = F(\phi(u)-), \qquad G(u+) = F(\phi(u)+).$$

At $u = c, d$ the endpoint conventions give these same identities for the relevant decreed limits. Subtracting proves $\Delta G(u) = \Delta F(\phi(u))$.

Match the upper and lower sums.

Take any $Q = \{c = u_0 < \cdots < u_n = d\}$ and its image partition $P = \{a = \phi(u_0) < \cdots < \phi(u_n) = b\}$. Bijection on the open subintervals gives

$$\sup_{(u_{k-1}, u_k)} g = \sup_{(\phi(u_{k-1}), \phi(u_k))} f,$$

and the analogous identity for infima. The one-sided-limit identities give matching interval weights, and $g(u_k) = f(\phi(u_k))$ gives matching jump terms. Thus

$$U_G(g, Q) = U_F(f, P), \qquad L_G(g, Q) = L_F(f, P).$$

Every partition of $[a, b]$ arises this way, by applying ϕ^{-1} to its points. Consequently the infima of the upper sums agree, and the suprema of the lower sums agree. Since f is F-integrable, those two common numbers coincide.

Conclusion. g is G-integrable, and $\displaystyle\int_c^d g\, dG = \int_a^b f\, dF$.

35.12 Exercise 35.12: A strictly increasing function with dense jumps

What we must show. Analyze the sum of positive jumps placed at every rational point of $[a, b]$.

First address the wording in the supplied edition.

The printed exercise says that the c_j are "positive integers" and that $\sum_j c_j < \infty$. These conditions are incompatible: each partial sum is at least its number of terms, so the series would diverge to $+\infty$. Thus there is no example satisfying the literal hypotheses. The intended, nonvacuous version uses *positive real numbers* with finite sum, just as in Example 3. We solve that version explicitly; for instance $c_j = 2^{-j}$ is an allowed choice. Let (u_j) enumerate the rationals in $[a, b]$ without repetition, and retain Ross's special convention for J_a.

(a) Show that the sum exists and increases strictly.

Since $0 \le J_{u_j}(t) \le 1$,

$$0 \le \sum_{j=1}^{N} c_j J_{u_j}(t) \le \sum_{j=1}^{\infty} c_j < \infty.$$

So the nondecreasing partial sums have a finite limit $F(t)$ at every point. Also their tails are at most $\sum_{j>N} c_j$, independently of t, so the convergence is uniform. For $a \le s < t \le b$, density of $\mathbb{Q}$ supplies a rational $u_m \in (s, t)$. The function J_{u_m} is 0 at s and 1 at t. Every other summand is increasing, hence has nonnegative increment. Subtracting the convergent series gives

$$F(t) - F(s) = \sum_{j=1}^{\infty} c_j [J_{u_j}(t) - J_{u_j}(s)] \ge c_m > 0.$$

This proves strict increase, not merely nondecrease.

(b) Prove continuity at every irrational point.

Let $t \in [a, b] \setminus \mathbb{Q}$ and $\varepsilon > 0$. Choose N with $\sum_{j>N} c_j < \varepsilon$. Because t differs from each of the finitely many $u_1, \ldots, u_N$, there is $\delta > 0$ such that $|x - t| < \delta$ crosses none of these points. Each of the first N step functions has the same value at x and t. For the remaining terms, $\left|J_{u_j}(x) - J_{u_j}(t)\right| \le 1$. Thus for $x \in [a, b]$ with $|x - t| < \delta$,

$$|F(x) - F(t)| \le \sum_{j>N} c_j \left|J_{u_j}(x) - J_{u_j}(t)\right| \le \sum_{j>N} c_j < \varepsilon.$$

This proves relative continuity at t, including an irrational endpoint if one occurs.

Show discontinuity at every rational point.

If $u_m \in (a, b]$, then $J_{u_m}(u_m) = 1$, whereas $J_{u_m}(x) = 0$ for every $x < u_m$. Since every summand is increasing, $F(u_m) - F(x) \ge c_m$ for $x < u_m$ in $[a, b]$. Points of the domain can approach u_m from the left while this gap stays at least $c_m > 0$, so F is discontinuous there. If the rational point is $u_m = a$, Ross defines $J_a(a) = 0$ and $J_a(x) = 1$ for $x > a$. Thus $F(x) - F(a) \ge c_m$ for all $x > a$, which proves right discontinuity at a. In fact Example 3 shows that the jump at u_m is exactly c_m.

Conclusion. Under the intended summable positive-real-weight hypothesis, F is strictly increasing, continuous exactly at irrational points of $[a, b]$, and discontinuous at every rational point. Under the printed positive-integer hypothesis, no such weight sequence exists.